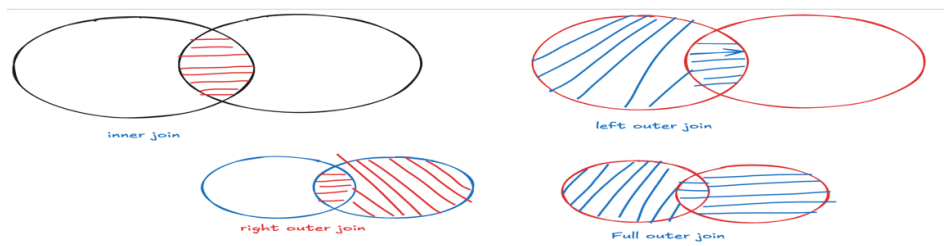


Joins in database



Joins in SQL

1. when in the output you need to get data from multiple tables, then we use joins
2. there are 3 types of joins, out of these inner join and outer join are most commonly used

cross join	it combines every row of one table with every row in another table
inner join	inner joins will always give you matching rows. these are of 3 types. 1. equi join → if the join condition is based on = operator then it is called as equi join 2. non equi join → if the join condition is based on operator other than = operator then it is called as non equi join. 3. self join → when you combine a table with itself then it is called as self join
outer join	outer join always gives you matching and non matching rows there are 3 type of outer join 1. left outer join → if you want to get matching rows and non matching rows from left side table then it is left outer join 2. right outer join → if you want to get matching rows and non matching rows from right side table then it is right outer join 3. full outer join → if you want to get matching rows and non matching rows from both side table then it is full outer join, to take full outer join in mysql we need to take union, but in oracle we can use full join

1. display all employees empno,ename,deptno,dname

<pre>select empno,ename,d.deptno,dname from emp e,dept d where e.deptno=d.deptno</pre>	<pre>select empno,ename,deptno,dname from emp e inner join dept d on e.deptno=d.deptno</pre>
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2. display all employees empno,ename,sal and grade

<pre>select empno,ename,sal,grade,losal,hisal from emp e, salgrade s where sal between losal and hisal;</pre>	<pre>select empno,ename,sal,grade,losal,hisal from emp e inner join salgrade s on sal between losal and hisal;</pre>
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3. display all employees empno,ename,deptno,dname who works in accounting department

<pre> select empno,ename,deptno,dname from emp e,dept d where e.deptno=d.deptno and d.dname='accounting'; </pre>	<pre> select empno,ename,deptno,dname from emp e inner join dept d on e.deptno=d.deptno where d.dname='accounting'; </pre>
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Select all employee who are working in accounting <pre> Select empno,ename,deptno,sal,grade From emp Where deptno = (select deptno From dept Where dname='Accounting') </pre>	Display employeeeno, name and department name <pre> Select empno,ename,d.deptno,dname From emp e, dept d Where e.deptno=d.deptno </pre>
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4. to display all employees with department name for all employees who works as CLERK

<pre> select empno,ename,dname from emp e, dept d where e.deptno=d.deptno and job='CLERK' </pre>	<pre> select empno,ename,dname from emp e inner join dept d on e.deptno=d.deptno where job='CLERK' </pre>

Display productid,pnameqty and price, cname,cdesc

<pre> select pid,pname,qty,price,cname,cdesc from product p , category c where p.catid=c.cid; </pre>	<pre> select pid,pname,qty,price,cname,cdesc from product p inner join category c on p.catid=c.cid; </pre>
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1. display course id, course name, faculty name and course duration from course and faculty table

<pre> Select cid,cname,duration,fname From course c,faculty f Where c.fid=f.facid </pre>	<pre> Select cid,cname,duration,fname From course c inner join faculty f on c.fid=f.facid </pre>
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2. display courseid,cname,roomname, duration of the course

<pre> Select cid,cname,duration,rname </pre>	<pre> Select cid,cname,duration,rname </pre>	
--	--	--

From course c,room r Where c.rid=r.roomid	From course c inner join room r on c.rid=r.roomid	
--	--	--

5. to display course id, course name,duration, room name and faculty name assigned to the course , for all courses with capacity >100

Select cid,cname,duration,rname,fname From course c,room r,faculty f Where c.rid=r.roomid and c.fid=f.facid and capacity > 100	Select cid,cname,duration,rname,fname From course c join room r on c.rid=r.roomid inner join faculty f On c.fid=f.facid Where capacity > 100
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6. write a query to display cid,cname, room name, also display rooms which are not assigned to the course

Select cid,cname,rname From course c right join room r On c.rid=r.roomid;	Select cid,cname,rname From room r left join course c On c.rid=r.roomid;
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7. write a query to display cid,cname, faculty name, also display faculties which are not assigned to the course

Select cid,cname,fname From course c right join faculty f on c.fid=f.facid;	Select cid,cname,fname From faculty f left join course c on c.fid=f.facid;
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8. display cid,cname,rid, rname from course and room table, also display courses for which no room is assigned, and rooms which not assigned to the course;

select cid,cname,rid,roomid,rname from course c left join room r on c.rid=r.roomid union select cid,cname,rid,roomid,rname from course c right join room r on c.rid=r.roomid;

9. display all course for which no room is assigned and also display rooms which are not assigned to course

Select cid,cname, null ,null from course Where rid is null union Select null,null,roomid,rname From room r Where not exists (select * from course c Where c.rid=r.roomid);	Select cid,cname,null roomid,null rname From course c left join room r on c.rid=r.roomid where rname is null union Select null, null,roomid,rname From course c right join room r on c.rid=r.roomid where cname is null;
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10. find all rooms which are not assigned to course and also display faculties which are not assigned to any course

Select roomid,rname,null facid,null name From room r left join course c On c.rid=r.roomid Where cname is null Union Select null, null,facid,fname From faculty f left join course c On c.fid=f.facid Where cname is null;	Select roomid,rname, null facid,null fname From room r Where not exists (select * from course c Where c.rid=r.roomid) Union Select null,null,facid,fname From faculty f Where not exists (select * from course c Where c.fid=f.facid);
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11. to display all employees empno,ename,dname and also display employees for whom no dept is assigned

select empno,ename,e.deptno,dname from emp e left join dept d on e.deptno=d.deptno;	select empno,ename,e.deptno,dname from dept d right join emp e on e.deptno=d.deptno;
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12. to display all employees empno,ename,dname and also display departments in which no employee is assigned

select empno,ename,e.deptno,dname from emp e right join dept d on e.deptno=d.deptno;	select empno,ename,e.deptno,dname from dept d left join emp e on e.deptno=d.deptno;
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7. to display all employees empno,ename,dname and also display employees who are not assigned to any department, also display all department in which no employee exists

select empno,ename,e.deptno,dname from emp e right join dept d on e.deptno=d.deptno union select empno,ename,e.deptno,dname from emp e left join dept d on e.deptno=d.deptno;

3. course (cid,cname, duration,roomid,fid)
faculty(fid,fname,subject)

room(rid,rname,rloc)

rid	rname	rloc
100	mogra	1 st floor
101	lotus	1 st floor
102	jasmin	2 nd floor

faculty

fid	fname	subject
1	Madhura	JAVA
2	Kishori	Database
3	Srujana	.net

course

cid	cname	duration	roomid	fid
1000	DAC	4	100	1
1001	DBDA	4	101	null
1002	DTISS	6	null	null

1. display all faculties who are not assigned to any course and also display rooms which are not assigned to any course

```
select cname,fid,fname,null roomno,null roomname
from course c right join faculty f on c.fid=f.fid
where cname is null
union
select cname, null,null,rid,rname
from course c right join room r on c.roomid=r.rid
where cname is null
```

2. display cname, rname assigned to it, also display rooms which are available

```
select cname,rname
from course c right join room r on r.rid=c.roomid;
```

3. display cname, fname assigned to it, also display faculties who are available

```
select cname,fname
from course c right join faculty f on f.fid=c.fid
```

4. display all courses name, faculty name and roomname assigned to all courses

select cname,fname,rname from courses c,faculty f, room r where c.roomid=r.rid and c.fid=f.fid	select cname,fname,rname from courses c inner join faculty f on c.fid=f.fid inner join room r on c.roomid=r.rid
--	--

5. to display cname,rname assigned to each course

```
select cname,rname
from course c inner join room r on
c.roomid=r.rid
```

```
select cname,rname
from course c , room r where
c.roomid=r.rid
```

6. to display cname, fname, course duration for all courses

```
select cname,fname,duration
from course c inner join faculty f
on c.fid=f.fid
```

```
select cname,fname,duration
from course c , faculty f
where c.fid=f.fid
```

7. write query to display empno,ename, dname, grade for all analysts and managers

8. write query to display empno,ename, dname, grade for all employees who works in sales department

```
Select empno,ename,dname,grade
From emp e inner join dept d on e.deptno=d.deptno inner join salgrade s
On e.sal between s.losal and s.hisal
Where dname ='sales'
```

9. write a query to display pairs of employee, like 1st employees salary < 2nd employees sal

10. write query to display pair of products like 1st products mfgdate = 2nd products mfgdate

11. display all pairs of employees who work under same manager

12. display all grades, which are not assigned to any employee

```
Select empno,ename,grade,losal,hisal
From emp e right join salgrade s
On e.sal between s.losal and s.hisal
Where ename is null;
```

13. display all faculties who are not assigned to any course and their sp_sub is java

```
Select facid,fname
From faculty f left join course c
On c.fid=f.facid
Where cname is null and f.sp_sub='java'
```

14. display all rooms which are not assigned to any course and are at 2nd floor

```
Select roomid,rname
From room r left join course c on c.rid=r.roomid
```

Where cname is null and rloc='2nd floor'